

KahfNews Telegram Bot

An automated Telegram bot that scrapes content from Kahf websites, generates social media posts using AI, and auto-posts to the @kahfnews channel in Bengali and English.

Project Overview

The KahfNews Telegram Bot is a sophisticated automation tool designed to streamline social media management for the Kahf ecosystem. It systematically extracts data from official web sources, including `kahf.com.tr` and `kahfguard.com`, to ensure that the content remains grounded in the latest company developments and product offerings. By leveraging the OpenAI GPT-4o model, the bot generates contextually relevant and engaging social media posts that resonate with a diverse audience.

Feature	Description
Web Scraping	Automated extraction of content from primary Kahf domains.
AI Content Generation	Utilization of advanced language models for high-quality post creation.
Bilingual Support	Generation of content in both Bengali and English to serve a global community.
Automated Scheduling	Precision posting at 9:00 AM and 7:00 PM BDT using APScheduler.

Technical Architecture

The system is built on a robust Python-based stack, ensuring reliability and ease of maintenance. The integration of `python-telegram-bot` facilitates seamless communication with the Telegram API, while `APScheduler` manages the temporal logic required for consistent posting. Data acquisition is handled by `BeautifulSoup4`, which parses HTML content into structured text suitable for AI processing.

Component	Technology
Language	Python 3.11
Bot Framework	python-telegram-bot
Scheduling	APScheduler

Data Parsing	BeautifulSoup4
AI Engine	OpenAI API

Deployment on Railway

1. Prepare your repository:

- Push this code to a GitHub repository.

2. Create a new project on Railway:

- Go to [Railway.app](https://railway.app).
- Click "New Project" -> "Deploy from GitHub repo".
- Select your repository.

3. Configure Environment Variables: Navigate to the **Variables** tab in the Railway dashboard and define the following environment variables to ensure secure and functional operation:

Variable	Description	Default / Example
<code>BOT_TOKEN</code>	Unique identifier for the Telegram bot.	<code>8944689234:AAGKlgwOdYvxOfx5XQO NLgqy3r9zszTPJ_g</code>
<code>CHANNEL_ID</code>	The target Telegram channel for posts.	<code>@kahfnews</code>
<code>OPENAI_API_KEY</code>	Authentication key for OpenAI services.	<code>your_api_key_here</code>
<code>OPENAI_API_BASE</code>	Endpoint for OpenAI API requests.	<code>https://api.openai.com/v1</code>

1. Deploy:

- Railway will automatically detect the `Procfile` and start the bot as a worker.

Local Development

1. Clone the repository.
2. Install dependencies: `pip install -r requirements.txt` .
3. Set the environment variables.

4. Run the bot: `python main.py` .